



# Suman Kumar Raj

Ocean Engineering and Naval Architecture  
Indian Institute Of Technology, Kharagpur

+91-9798196180

sumanraj4176@gmail.com

LinkedIn | GitHub | Portfolio

## EDUCATION

| Degree/Certificate | Institute                                   | CGPA/Percentage | Year      |
|--------------------|---------------------------------------------|-----------------|-----------|
| B.TECH + M.TECH    | Indian Institute of Technology, Kharagpur   | 8.12            | 2020–2025 |
| Class XII (CBSE)   | S K P Vidya Vihar Mandroza Bhagalpur, Bihar | 92%             | 2020      |
| Class X (CBSE)     | D P S Sabour Bhagalpur, Bihar               | 10              | 2018      |

## PROJECTS

- **Multi-Agent Research Assistant** [↗](#)  
*Python, LangGraph, LangChain, OpenAI, Tavily, Wikipedia, LangSmith*
  - o Built a multi-agent research system using **LangGraph** to decompose complex topics into specialized analyst perspectives with human-in-the-loop refinement.
  - o Implemented parallel information retrieval using **Tavily** and **Wikipedia**, enabling agents to gather and synthesize context from multiple sources.
  - o Implemented a map-reduce pipeline to parallelize analyst interviews and synthesize a comprehensive final report.
- **MyTaskManager** [↗](#)  
*Python, LangGraph, LangChain, OpenAI, Pydantic, Trustcall*
  - o Built a stateful AI task-management agent using LangGraph that maintains short-term conversational state and structured long-term memory across interactions.
  - o Designed separate **Profile**, **To-Do**, and **Instruction** memory with Pydantic schemas, enabling structured memory extraction, updates, and personalized task management.
  - o Implemented conditional agent routing and cyclic workflows to determine when memory updates are required, route operations to the appropriate memory node, and return control to the main agent.
- **Fingerprint Liveness Detection (Presentation Attack Detection)** [↗](#)  
*Python, TensorFlow, Keras, MobileNetV3-Small, OpenCV*
  - o Built a **Presentation Attack Detection (PAD)** system using **MobileNetV3-Small** transfer learning to classify fingerprint images as **LIVE** or **SPOOF** for biometric authentication.
  - o Developed an end-to-end pipeline covering 70/15/15 dataset splitting, image augmentation, ImageNet normalization, model training, threshold calibration, and single-image inference.
  - o Calibrated the threshold against a target **BPCER** of **3%** and evaluated held-out test performance using **APCER**, **BPCER**, **ACER**, **EER**, **ROC-AUC**, **F1-score**, and **APCER–BPCER** trade-off analysis.
- **LendingClub Risk Prediction** [↗](#)  
*Python, Pandas, NumPy, Scikit-learn, TensorFlow/Keras*
  - o Built an end-to-end credit risk prediction pipeline using TensorFlow/Keras on **396K+** LendingClub loan records, covering missing-value handling, categorical encoding, feature engineering, and MinMax scaling.
  - o Trained a deep neural network with ReLU activations, dropout, & sigmoid output to classify loans as fully paid or defaulted.
  - o Achieved a **0.93 F1-score** on the test set, effectively handling significant class imbalance in loan default prediction.

## SKILLS SUMMARY

- **Generative AI:** LangChain, LangGraph, Agentic AI, Memory Systems, Multi-Agent Systems, LLM Orchestration, RAG, VectorDB, GraphDB, Prompt Engineering, Tool & Function Calling, Structured Outputs, LLM Evaluation.
- **Machine Learning:** Classification, Regression, EDA, NLP, Neural Networks.
- **Programming Languages:** Python, SQL
- **Frameworks and Libraries:** TensorFlow, Keras, PyTorch, Scikit-learn, **NumPy**, **Pandas**, **Matplotlib**, **Seaborn**.
- **Platforms & Tools:** Git, GitHub, LangSmith, VS Code, Jupyter Notebook, Google Colab, Kaggle.
- **Data Structures and Algorithms, DBMS, Object-Oriented Programming, MLOps, Probability and Statistics**

## EXPERIENCE

- **AI/ML Engineer Intern | MakeMyBrain** [May'24 – Jul'24]
  - o Built an end-to-end **mood-based music recommendation RAG system**, processing structured metadata, acoustic features, and textual profiles across audio catalogs.
  - o Engineered a **hybrid search pipeline** combining **BM25 keyword matching** and **dense vector embeddings** ( $\alpha = 0.4$ ), boosting retrieval **Recall@10** from **0.61** to **0.79** compared to lexical search alone.
  - o Implemented a **cross-encoder re-ranking** stage to filter **top 100** retrieved candidates down to **top 10–20 tracks** before **LLM generation**, ensuring personalized and context-aware recommendations.
- **Mentor | AlgoZenith** [Oct'22 – May'24]
  - o Mentored **500+** students in DSA, guiding them through complex coding challenges and technical problem-solving.
- **Teaching Assistant | Indian Institute of Technology, Kharagpur** [Aug'23 - May'25]
  - o Conducted tutorials, evaluated assignments, and led doubt-clearing sessions for undergraduate courses at IIT Kharagpur.

## AWARDS AND ACHIEVEMENTS

- Qualified **GATE 2024** with All India Rank (AIR) **64**.
- Received the prestigious **Prof. J.P. Ghose Memorial Award (2022–23)** for exceptional academic performance.
- **Gold Medal** recipient and distinguished participant in the Gymkhana Championship Choreography (2021–2022).

## OPEN SOURCE

- **CodeBeat — Open-Source Python Package | PyPI** | [↗](#)
  - o Created a Python profiling package for function tracing and execution-time analysis to identify performance bottlenecks.